

Notes: Morchio & Moser (AER, 2026)

The Gender Pay Gap: Micro Sources and Macro Consequences

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Linked employer–employee data	2
2.2	Gender-specific employer pay	3
2.3	Between- and within-employer pay gaps	3
2.4	Workplace amenities and total compensation	3
3	Model and empirical specifications	3
3.1	Workers, search, and employment	3
3.2	Firms and endogenous pay, amenities, and hiring	4
3.3	Decomposing pay dispersion	4
3.4	Identification logic	4
4	Identification and empirical design	4
4.1	What identifies employer pay gaps	4
4.2	What identifies amenities	5
4.3	What identifies counterfactual policy effects	5
5	Tables and figures	5
5.1	Table 1, Figure 1, and Tables 2–3: Empirical gender pay gaps	5
5.2	Tables 4–7: Estimated labor-market objects and firm characteristics	6
5.3	Tables 8–9 and Figure 2: Cost elasticities and model fit	7
5.4	Figures 3–7 and Table 10: Pay, amenities, ranks, and compensation	8
5.5	Tables 11–13: Gender gaps and counterfactual policies	9
6	Short summary	10
7	Conclusion	10
7.1	Core conclusion	10
7.2	Economic intuition	10
7.3	Takeaway	11

1 Big picture

- **Question.** What microeconomic forces generate the gender pay gap, and what are their aggregate consequences for output, employment, and welfare?
- **Setting.** The paper uses linked employer–employee data from Brazil for 2007–2014, a setting with a large raw gender earnings gap and rich worker, employer, occupation, hours, tenure, and education information.
- **Reduced-form starting fact.** Brazil has a mean log gender earnings gap of 13.3 log points. After controlling for worker heterogeneity with gender-specific AKM employer effects, the employer component of the gap is 11.2 log points.
- **Sorting result.** Most of the employer-pay gap is *between employers*: women work disproportionately at lower-paying employers. The between-employer component is 8.9 log points, or 79.5 percent of the employer fixed-effect gap.
- **Structural contribution.** The paper develops an equilibrium search model with endogenous firm pay, workplace amenities, hiring, and firm size. Men and women can face different ladders because firms differ in productivity, amenity costs, and gender wedges.
- **Identification contribution.** The model is rich but constructively identified from linked employer–employee data. The authors recover gender-specific firm pay, utility ranks, amenity valuations, hiring intensities, productivity, gender wedges, and economy-wide cost elasticities.
- **Main structural result.** Amenities are quantitatively important. Employer pay dispersion mostly reflects compensating differentials rather than true utility dispersion.
- **Gender compensation result.** Women have lower pay but higher amenity valuations. The gender gap in total compensation is only 3.5 log points, about 31.6 percent of the pay gap.
- **Policy result.** Equal pay and equal hiring policies reduce measured pay gaps but do not improve output or worker welfare in the estimated equilibrium. Firms and workers adjust through amenities, employment, and sorting.
- **Takeaway.** A large pay gap does not automatically imply an equally large utility gap. But the model also implies that closing pay gaps by policy can be costly if it distorts the equilibrium allocation of pay, amenities, and hiring.

2 Data and measurement

2.1 Linked employer–employee data

- The main data source is RAIS, Brazil’s matched employer–employee administrative dataset, over 2007–2014.
- The final sample contains 267.3 million worker-year observations, 56.3 million unique workers, and 607,106 unique employers.
- The sample is 38.2 percent female. Women are more educated on average but earn less and work fewer contractual hours.
- Earnings are real monthly earnings; the main outcome is log real monthly earnings.

Interpretation: The data allow the authors to separate gender differences in worker characteristics from gender differences in employer pay, sorting, and amenities.

2.2 Gender-specific employer pay

The paper estimates a gender-specific AKM-style earnings equation:

$$\ln w_{ijt} = \alpha_i + \psi_{G(i)j} + X_{it}\beta_{G(i)} + \varepsilon_{ijt}, \quad (1)$$

where α_i is a worker fixed effect, $\psi_{G(i)j}$ is a gender-specific employer fixed effect, and X_{it} includes controls such as hours, occupation, tenure, experience, and education-age variables.

Interpretation: The key empirical object is not just whether women earn less at the same firm, but whether they work at firms that pay lower employer premiums.

2.3 Between- and within-employer pay gaps

The gender gap in employer fixed effects is decomposed as

$$\underbrace{E[\psi_{MJ(i,t)} \mid G(i) = M] - E[\psi_{MJ(i,t)} \mid G(i) = F]}_{\text{between-employer gap}} + \underbrace{E[\psi_{MJ(i,t)} - \psi_{FJ(i,t)} \mid G(i) = F]}_{\text{within-employer gap}}. \quad (2)$$

Interpretation: The between component asks whether men and women work at different employers. The within component asks whether the same employer pays men and women differently after worker controls.

2.4 Workplace amenities and total compensation

Workers receive utility from monetary pay and amenities:

$$x = w + \beta_g a, \quad (3)$$

where w is pay, a is a firm amenity, and β_g is the gender-specific value of that amenity. The paper often writes amenity valuations relative to pay as

$$\tilde{a} = x/w, \quad \ln w + \ln \tilde{a} = \ln x. \quad (4)$$

Interpretation: Pay is only one component of compensation. A firm can be low-pay but high-amenity, and workers may rationally sort toward such firms.

3 Model and empirical specifications

3.1 Workers, search, and employment

- Workers are infinitely lived, risk neutral, and differ by gender $g \in \{M, F\}$ and ability z .
- Workers can be nonemployed or employed and receive random job offers both from nonemployment and on the job.
- Search is segmented by worker type, so men and women can face different arrival rates, destruction rates, reservation values, and firm ladders.
- Workers rank jobs by total utility x , not only by pay w .

Interpretation: This is why a job-to-job transition can involve a pay cut: the worker may move to a job with higher amenity valuation or utility.

3.2 Firms and endogenous pay, amenities, and hiring

- Firms differ in productivity p , gender wedges τ , and gender-specific amenity cost shifters.
- Firms choose pay, amenities, and vacancies/hiring to maximize profits.
- Vacancy costs are convex; amenity costs are also governed by an elasticity parameter.
- A gender wedge can be interpreted as taste-based discrimination or a firm-level comparative advantage in employing one gender.

Interpretation: Firm pay is not taken as exogenous. It is an equilibrium object jointly determined with amenities, recruiting, and worker sorting.

3.3 Decomposing pay dispersion

The model implies a decomposition of employer pay dispersion into utility and amenities. With $\ln w + \ln \tilde{a} = \ln x$,

$$\text{Var}(\ln w) = \text{Var}(\ln x) + \text{Var}(\ln \tilde{a}) - 2 \text{Cov}(\ln x, \ln \tilde{a}). \quad (5)$$

Interpretation: The variance of pay across employers can reflect true utility differences or compensating differentials for amenities. The paper finds the latter dominates.

3.4 Identification logic

- The AKM equation delivers gender-specific employer pay components.
- Worker flows and job-to-job transitions reveal utility rankings because workers move toward higher utility offers.
- Comparing utility offers with pay recovers amenity valuations.
- Comparing equilibrium outcomes for men and women at the same employer identifies gender wedges.
- Aggregate moments identify the economy-wide vacancy and amenity cost elasticities.

Interpretation: The model is identified without relying on strong parametric assumptions about the distribution of firm types.

4 Identification and empirical design

4.1 What identifies employer pay gaps

- Worker fixed effects absorb time-invariant worker heterogeneity.
- Gender-specific employer fixed effects allow the same employer to have different pay components for men and women.
- Leave-out methods address limited-mobility bias in variance decompositions.
- The connected-set structure ensures employer effects are comparable across employers linked by worker mobility.

Interpretation: The reduced-form evidence is not simply a raw wage comparison. It uses worker mobility to estimate firm pay premiums for otherwise comparable workers.

4.2 What identifies amenities

- Pay alone does not reveal utility because workers may value nonpay job attributes.
- Job-to-job moves reveal preferences: workers accept an offer when the utility value is high enough.
- Once utility ranks are recovered, the difference between utility and pay identifies amenity valuations.

Interpretation: The key revealed-preference idea is that observed mobility patterns contain information about how workers trade pay against amenities.

4.3 What identifies counterfactual policy effects

- Policy experiments are evaluated in equilibrium, not by holding firm behavior fixed.
- When pay or hiring constraints change, firms adjust vacancies, amenities, and employment.
- The policy results therefore combine distributional effects with general-equilibrium responses.

Interpretation: Equal treatment policies can reduce measured gender gaps while reducing output or welfare if they distort firms' equilibrium choices and worker sorting.

5 Tables and figures

5.1 Table 1, Figure 1, and Tables 2–3: Empirical gender pay gaps

Read:

- Table 1 shows a 13.3 log point raw gender earnings gap in Brazil during 2007–2014.
- Figure 1 shows that men's employer fixed effects are shifted to the right of women's, and that within-employer gender differences are centered near a positive male advantage.
- Table 2 shows that employer fixed effects explain 13.0 percent of men's plug-in earnings variance and 11.5 percent of women's.
- Table 3 decomposes the 11.2 log point gender gap in employer FEs into an 8.9 log point between-employer component and a 2.3 log point within-employer component.

Economic message: Most of the gender pay gap reflects sorting across employers, not only unequal pay within the same employer.

	Overall	Men	Women
Mean log real monthly earnings (standard deviation)	7.211 (0.693)	7.262 (0.697)	7.129 (0.679)
Mean years of education (standard deviation)	11.1 (3.3)	10.4 (3.3)	12.1 (2.9)
Mean years of age (standard deviation)	33.6 (9.4)	33.5 (9.4)	33.8 (9.4)
Mean employer size (standard deviation)	2,815 (16,418)	1,774 (11,510)	4,496 (22,059)
Mean contractual work hours (standard deviation)	41.7 (5.1)	42.6 (3.9)	40.3 (6.4)
Mean years of tenure (standard deviation)	3.9 (5.6)	3.6 (5.2)	4.5 (6.1)
Share non-White	0.378	0.409	0.327
Share female	0.382		
Mean log gender earnings gap	0.133		
Number of worker-years	267,312,672	165,144,176	102,168,504
Number of unique workers	56,297,216	33,761,352	22,535,864
Number of unique employers	607,106	402,880	204,226

Notes: This table reports summary statistics for workers in the final sample, separately for the overall population, for men only, and for women only. Since information on race is missing for a significant number of observations, conditional means are reported for the share of non-White workers.

Source: RAIS (2007–2014)

(a) Table 1: summary statistics

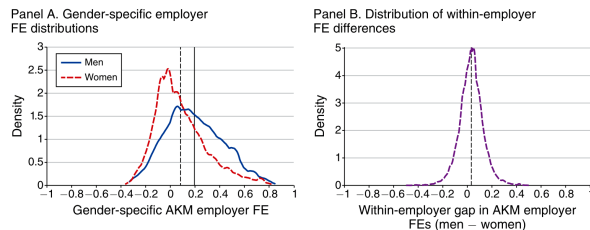


FIGURE 1. PREDICTED AKM EMPLOYER FEs FOR WOMEN AND MEN

Notes: This figure shows kernel density plots of estimated gender-specific employer FEs based on estimating the

(b) Figure 1: gender-specific AKM employer FEs

	Plug-in	Leave-out	Plug-in	Leave-out
Variance of log earnings	0.497	0.258	0.482	0.250
Variance components:				
Employer fixed effects (percent)	0.065 (13.0%)	0.064 (25.0%)	0.056 (11.5%)	0.055 (22.2%)
Person fixed effects (percent)	0.116 (23.3%)	0.097 (37.7%)	0.117 (24.3%)	0.099 (39.6%)
Correlation	0.212	0.245	0.255	0.297
R^2	0.921	0.777	0.929	0.793
Mean employer fixed effects	0.195	0.195	0.083	0.083

Notes: This table shows the variance components of log earnings based on equation (1). The variance components correspond to the variance decomposition $\text{Var}(\ln w_{it}) = \text{Var}(\alpha_i) + \text{Var}(\psi_{cit}) + \text{Var}(\tilde{X}_{it}\beta_{cit}) + 2\sum \text{Cov}(\cdot) + \text{Var}(\varepsilon_{it})$. The “plug-in” columns refer to conventional plug-in estimates, while the “leave-out” columns refer to estimates that correct for limited-mobility bias, following KSS. The variances shown in the “plug-in” columns are the variances of raw log earnings, while those in the “leave-out” columns are the variances of residualized log earnings conditional on the same controls as those in equation (1).

Source: RAIS (2007–2014)

(a) Table 2: variance decomposition

TABLE 3—KITAGAWA-OAXACA-BLINDER DECOMPOSITIONS OF THE GENDER GAP IN EMPLOYER FES

Gender gap in employer fixed effects	Between-employer gap		Within-employer gap	
	Level	Share (%)	Level	Share (%)
0.112	0.089	79.5	0.023	20.5

Notes: This table shows results from the Kitagawa-Oaxaca-Blinder decomposition of the gender gap in employer fixed effects into a between-employer gap and a within-employer gap. The decomposition corresponds to equation (2) and uses men’s employer fixed effects for computing the between-employer component. An alternative decomposition using women’s employer FEs for computing the between-employer component is presented in Table B.1 in Supplemental Appendix B.2.

(b) Table 3: KOB decomposition

5.2 Tables 4–7: Estimated labor-market objects and firm characteristics

Read:

- Table 4 estimates monthly offer arrival, job destruction, on-the-job offer, and nonemployment flow values separately by gender.
- Men receive more total job offers than women; women have lower job destruction rates.
- Table 5 validates the model’s firm productivity measure using log revenue per worker; the coefficient is 0.898.
- Table 6 relates gender wedges to employer characteristics; female managers reduce estimated gender wedges, while larger size and longer work hours raise them.
- Table 7 relates amenity valuations to observed workplace characteristics; flexibility, parental leave, stability, and size are connected to amenities differently by gender.

Economic message: The structural objects have empirical content: estimated productivity, wedges, and amenities line up with observable firm features.

TABLE 4—JOB OFFER ARRIVAL RATES, JOB DESTRUCTION RATES, AND FLOW VALUES OF NONEMPLOYMENT

Parameter	Description	Men	Women
μ_g	Population shares	0.599	0.401
λ_g^U	Offer arrival rate from nonemployment	0.104	0.091
δ_g	Job destruction rate	0.036	0.028
s_g^E	Relative arrival rate of voluntary on-the-job offers	0.090	0.077
s_g^G	Relative arrival rate of involuntary on-the-job offers	0.101	0.082
b_g	Flow value of nonemployment	2.278	2.225

Notes: This table shows the estimated values of all labor market parameters—specifically, the offer arrival rate from nonemployment λ_g^U , the job destruction rate δ_g , the relative arrival rate of voluntary on-the-job offers s_g^E , the relative arrival rate of involuntary job offers s_g^G , and the flow value of nonemployment b_g —separately by gender $g \in \{M, F\}$. All rates are monthly.

Source: Model estimates based on RAIS (2007–2014).

(a) Table 4: labor-market parameters

TABLE 5—REGRESSING EMPIRICAL PRODUCTIVITY MEASURES ON MODEL ESTIMATES OF FIRM PRODUCTIVITY

	Coefficient	(SE)
Revenue per worker	0.898	(0.155)
R^2	0.120	

Notes: This table reports the estimated coefficient from regressing empirical revenue per worker on structurally estimated firm productivity p —see equation (24). Details of the variable construction are presented in Supplemental Appendix E.2. Standard errors are clustered at the employer level.

Source: Model estimates based on RAIS (2007–2014).

(b) Table 5: productivity validation

	Coefficient	(SE)
Female manager	0.007	(0.002)
Nonroutine manual task intensity	0.002	(0.007)
Nonroutine interpersonal task intensity	-0.002	(0.006)
Mean working hours	-0.016	(0.005)
No major financial stakeholders	-0.009	(0.002)
log size	-0.158	(0.007)
R^2	0.628	
Within- R^2	0.090	

Notes: This table reports estimated coefficients from regressing log transformed gender wedges, $\ln(1 - \tau)$, on observable employer characteristics—see equation (25). Estimates are conditional on municipality and sector FEs. All covariates are standardized and expressed in z-scores relative to the population of all firms for each gender. Details of all covariates are presented in Supplemental Appendix E.3. Standard errors are clustered at the employer level.

Source: Model estimates based on RAIS (2007–2014).

(a) Table 6: gender wedges

	Men		Women	
	Coefficient	(SE)	Coefficient	(SE)
Part-time work incidence	-0.006	(0.013)	0.009	(0.007)
Working hours flexibility	0.008	(0.013)	0.021	(0.006)
Parental leave generosity	0.087	(0.024)	0.023	(0.008)
Income fluctuations	-0.036	(0.037)	0.000	(0.007)
Workplace hazards	0.017	(0.015)	-0.002	(0.006)
Incidence of unjust firings	-0.031	(0.014)	-0.025	(0.009)
Incidence of workplace deaths	-0.029	(0.015)	-0.062	(0.010)
log size	0.206	(0.018)	0.152	(0.021)
R^2	0.691		0.447	
Within- R^2	0.233		0.117	

Notes: This table reports estimated coefficients from regressing log amenity valuations, $\ln(\beta_{ag})$, on observable gender-specific employer characteristics—see equation (26). Estimates are conditional on municipality and sector FEs. All covariates are standardized and expressed in z-scores relative to the population of all firms for each gender. Details of all covariates are presented in Supplemental Appendix E.4. Standard errors are clustered at the employer level.

Source: Model estimates based on RAIS, 2007–2014.

(b) Table 7: amenity regressions

5.3 Tables 8–9 and Figure 2: Cost elasticities and model fit

Read:

- Table 8 estimates a vacancy cost elasticity of 2.062 and an amenity cost elasticity of 5.721.
- Table 9 shows that the model matches the main moments closely: the gender log pay gap, between/within gap, pay variances, job-to-job transitions, pay-cut probabilities, and correlation between men and women’s pay.
- Figure 2 shows that the model closely fits the distributions of log firm pay and log firm size by gender.

Economic message: The estimated equilibrium model is disciplined by both pay moments and worker-flow moments, not just by fitting the gender pay gap.

TABLE 8—ESTIMATES OF ECONOMY-WIDE PARAMETERS					
Elasticity	Cost function	Value	Moment	Data	Model
η^v	Vacancies	2.062	Slope of log pay on log value added	0.179	0.179
η^a	Vacancies	5.721	Cost share of amenities	0.08	0.08

Notes: This table reports the estimated values of the elasticity of the vacancy cost function η^v and the elasticity of the amenity cost function η^a . Targeted moments are the elasticity of pay with respect to value added per worker from Alvarez et al. (2018) and the cost share of amenities in value added based on Bieri, Kuminoff, and Pope (2023). See Supplemental Appendix E.1 for details on the construction of the aggregate statistics.

Source: Model estimates based on RAIS, 2007–2014.

(a) Table 8: economy-wide elasticities

Moment	Description	Data	Model
$\mathbb{E}[\psi_M - \psi_F]$	Gender log pay gap	0.112	0.107
$\mathbb{E}[\psi_F g = M] - \mathbb{E}[\psi_F g = F]$	Gender log pay gap between employers	0.089	0.082
$\mathbb{E}[\psi_F - \psi_M g = F]$	Gender log pay gap within employers	0.023	0.025
$\text{Var}[\psi_M]$	Variance of men’s pay	0.054	0.053
$\text{Var}[\psi_F]$	Variance of women’s pay	0.044	0.044
$\text{Var}[\psi_M - \psi_F]$	Variance of gender pay gap	0.009	0.010
$\mathbb{E}[\lambda_M^j(1 - F_M(x)) + \lambda_F^j]$	Job to job transition rate for men	0.013	0.015
$\mathbb{E}[\lambda_F^j(1 - F_F(x)) + \lambda_M^j]$	Job to job transition rate for women	0.010	0.011
$\mathbb{P}[\psi_M < \psi_M]$	Wage decline probability after job to job for men	0.416	0.480
$\mathbb{P}[\psi_F < \psi_F]$	Wage decline probability after job to job for women	0.430	0.501
$\text{Corr}(\psi_M, \psi_F)$	Correlation between men’s and women’s pay	0.921	0.955

Notes: This table reports the fit of the model in terms of data-based and model-based moments across genders $g \in \{M, F\}$. All statistics are employment-weighted.

Source: Model estimates based on RAIS, 2007–2014.

(b) Table 9: model fit

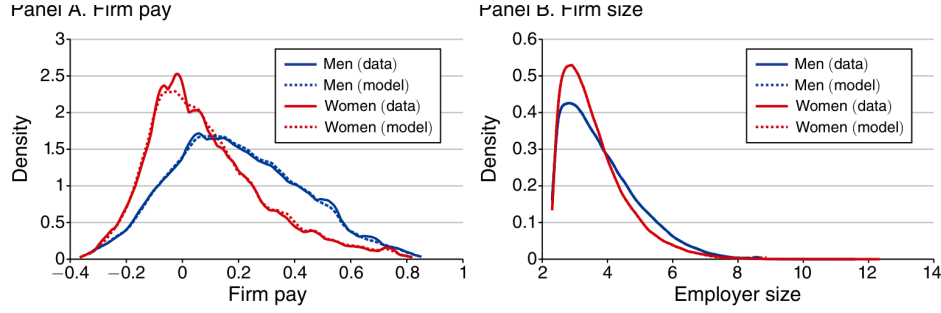


FIGURE 2. MODEL FIT IN TERMS OF THE DISTRIBUTIONS OF LOG FIRM PAY AND LOG FIRM SIZE, BY GENDER

Notes: This figure plots the model fit in terms of the data versus model distributions of log firm pay (panel A) and log firm size (panel B). Blue lines show the results for men, while red lines show the results for women. Solid lines show the data, while dashed lines show the model.

Source: Model estimates based on RAIS (2007–2014)

Figure 6: Figure 2: model fit for firm pay and firm size distributions

5.4 Figures 3–7 and Table 10: Pay, amenities, ranks, and compensation

Read:

- Figure 3 shows that for men, higher employer ranks tend to coincide with higher pay ranks. For women, high utility ranks are less tightly linked to pay ranks.
- Figure 4 shows a much stronger rank-amenity relation for women than for men.
- Figure 5 shows that women are more concentrated at employers with high amenity valuations.
- Figure 6 shows that men's total compensation rises mostly through pay along the employer ladder, while women's rises mostly through amenities.
- Figure 7 shows that amenity shares in total compensation average 48.9 percent for men and 52.7 percent for women.
- Table 10 decomposes employer pay dispersion: for men, 94.3 percent of pay dispersion is attributed to log amenities; for women, the amenity component exceeds 100 percent because the covariance term is negative.

Economic message: A large share of observed pay dispersion reflects compensating differentials for workplace amenities rather than pure utility dispersion.

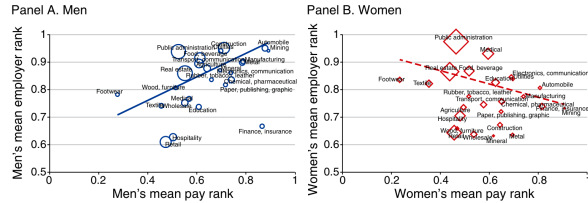


FIGURE 3. SECTORAL EMPLOYER RANKS AGAINST EMPLOYER PAY RANKS, BY GENDER

Notes: This figure shows ranks of employer utility (x_k) against ranks of employer pay (w_k) for men in panel A and for women in panel B across 25 sectors. Marker sizes represent employment weights on which the linear fit lines are based.

Source: Model estimates based on RAIS, 2007–2014.

(a) Figure 3: sectoral ranks versus pay ranks

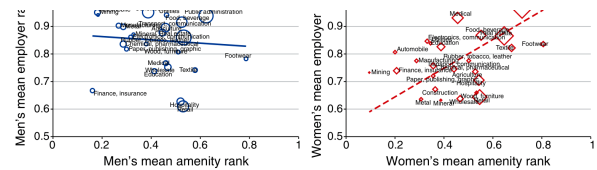


FIGURE 4. SECTORAL EMPLOYER RANKS AGAINST EMPLOYER AMENITY RANKS, BY GENDER

Notes: This figure shows ranks of employer utility (x_k) against ranks of employer amenity valuations ($\beta_k a_k$) for men in panel A and for women in panel B across 25 sectors. Marker sizes represent employment weights on which the linear fit lines are based.

Source: Model estimates based on RAIS (2007–2014).

(b) Figure 4: sectoral ranks versus amenity ranks

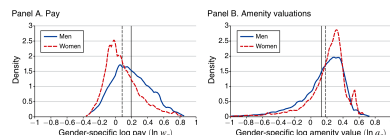


FIGURE 5. ESTIMATED DISTRIBUTIONS OF PAY AND AMENITY VALUATIONS, BY GENDER

Notes: This figure shows the gender-specific employment-weighted distributions of log pay ($\ln w_g$) in panel A and of log amenity valuations ($\ln a_g$) in panel B. The gray patterned vertical lines show the population means for the corresponding gender $g \in \{M, F\}$.

Source: Model estimates based on RAIS (2007–2014).

(a) Figure 5: pay and amenity distributions

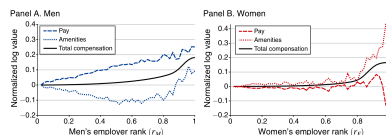


FIGURE 6. PAY, AMENITY VALUATIONS, AND TOTAL COMPENSATION ACROSS EMPLOYER RANK, BY GENDER

Notes: This figure shows the relative values of log pay ($\ln w_g$), log amenity valuations ($\ln a_g$), and log total compensation ($\ln c_g$) across firm ranks (r_g) separately for men in panel A and for women in panel B. All log values are normalized to zero for the gender-specific group of bottom-ranked employers.

Source: Model estimates based on RAIS (2007–2014).

(b) Figure 6: pay, amenities, and total compensation by rank

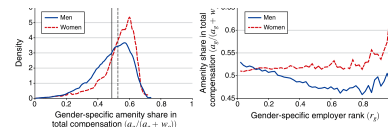


FIGURE 7. DISTRIBUTION OF AMENITY SHARES IN TOTAL COMPENSATION, BY GENDER

Notes: Panel A of this figure shows the gender-specific employment-weighted distribution of amenity shares in total compensation ($a_g/(w_g + a_g)$). The gray patterned vertical lines show the population means for the corresponding gender $g \in \{M, F\}$. Panel B of this figure shows the amenity share in total compensation ($a_g/(w_g + a_g)$) across employer ranks (r_g) separately by gender $g \in \{M, F\}$.

Source: Model estimates based on RAIS (2007–2014).

iv. As shown in Table 10, the variance of log employer nav is 0.054 for men and 0.044

(c) Figure 7: amenity shares

TABLE 10—DECOMPOSITION OF EMPLOYER PAY DISPERSION INTO UTILITY AND AMENITY TERMS

Variances	Men		Women	
	Level	Share (%)	Level	Share (%)
Variance of log pay	0.054		0.044	
Variance components:				
log utility	0.002	4.4	0.003	6.0
log amenities	0.051	94.3	0.046	105.0
Covariance between log utility and log amenities	0.001	1.3	−0.005	−11.0
Covariance components:				
Covariance between log utility and log pay	0.003	5.0	0.000	0.5
Covariance between log amenities and log pay	0.052	95.0	0.044	99.5

Notes: This table shows the variance and covariance components of log pay ($\ln w$). To this end, we define “amenities” as the utility-to-wage ratio $\tilde{a} = x/w$ so that $\ln w + \ln \tilde{a} = \ln x$. The variance components correspond to the variance decomposition $\text{Var}(\ln w) = \text{Var}(\ln x) + \text{Var}(\ln \tilde{a}) - 2\text{Cov}(\ln x, \ln \tilde{a})$. The covariance components correspond to the covariance decomposition $\text{Cov}(\ln w, \ln x) = \text{Cov}(\ln w, \ln \tilde{a}) - \text{Cov}(\ln w, \ln \tilde{a})$.

Source: Model estimates based on RAIS (2007–2014).

Figure 9: Table 10: employer pay dispersion decomposition

5.5 Tables 11–13: Gender gaps and counterfactual policies

Read:

- Table 11 shows that the gender pay gap is 11.2 log points, but the amenity-valuation gap is -7.7 log points in favor of women, so the total-compensation gap is only 3.5 log points.
- Eliminating firm heterogeneity in amenities cuts the gender pay gap from 10.7 to 5.4 log points, increases output by 1.6 percent, and increases welfare by 1.4 percent.
- Table 13 shows that equal pay and equal hiring policies substantially reduce the pay gap but lower output and worker welfare relative to the baseline.
- Equal pay reduces the pay gap to 2.8 log points but output falls to 0.986 and welfare to 0.996.
- Equal hiring reduces the pay gap to 3.2 log points but welfare falls to 0.992 and employment reallocates strongly from men to women.

Economic message: Counterfactuals reveal the central tension: amenities reduce the pay gap’s welfare interpretation, while equal-treatment policies can reduce measured gaps without improving welfare.

TABLE 11—KITAGAWA-OAXACA-BLINDER DECOMPOSITIONS OF GAPS IN PAY, AMENITY VALUATIONS, AND TOTAL COMPENSATION

	Gender gap	Between-employer gap		Within-employer gap	
		Level	Share (%)	Level	Share (%)
Pay	0.112	0.089	79.5	0.023	20.5
Amenity-valuation	-0.077	-0.087	113.5	0.010	-13.5
Total compensation	0.035	0.002	6.1	0.033	93.9

Notes: This table shows results from the Kitagawa-Oaxaca-Blinder decomposition of the gender gaps in log pay ($\ln w$), log amenity valuations ($\ln \bar{a} = \ln(x/w)$), and log total compensation ($\ln s$) into a between-employer and a within-employer gap. To this end, we define "amenity valuations" as the utility-to-wage ratio $\bar{a} = s/w$ so that $\ln w + \ln \bar{a} = \ln s$. All decompositions are based on equation (2). Table B.1 in Supplemental Appendix B.2 as well as Tables F.1–F.2 in Supplemental Appendix F.1 show alternative decompositions using men's compensation for computing the between-employer component of pay, amenity valuations, and total compensation, respectively.

Source: Model estimates based on RAIS (2007–2014).

(a) Table 11: pay, amenities, and total compensation gaps

TABLE 12—EFFECTS OF ELIMINATING FIRM HETEROGENEITY IN AMENITIES

	Baseline	Same amenities
Gender log pay gap	0.107	0.054
between employers	0.082	0.020
within employers	0.025	0.034
Gender log amenities gap	-0.068	-0.011
between employers	-0.075	-0.009
within employers	0.007	-0.002
Gender log utility gap	0.039	0.042
between employers	0.007	0.010
within employers	0.032	0.032
Output	1.000	1.016
Worker welfare	1.000	1.014
for men	1.000	1.014
for women	1.000	1.013
Total employment	0.771	0.783
for men	0.764	0.772
for women	0.781	0.799

Notes: This table reports results from equilibrium counterfactuals. Log amenities are defined as the log amenity valuations ($\ln \bar{a} = \ln(x/w)$). The baseline economy (column 0) is compared to a counterfactual without differences in amenities across employers (column 1).

Source: Model estimates based on RAIS (2007–2014).

(b) Table 12: eliminating amenity heterogeneity

TABLE 13—EFFECTS OF SIMULATED EQUAL PAY AND EQUAL HIRING POLICIES

	Baseline	Equal pay policy	Equal hiring policy
	(0)	(1)	(2)
Gender log pay gap	0.107	0.028	0.032
between employers	0.082	0.028	0.006
within employers	0.025	-0.000	0.026
Gender log amenities gap	-0.068	-0.000	0.010
between employers	-0.075	-0.027	-0.006
within employers	0.007	0.027	0.016
Gender log utility gap	0.039	0.027	0.042
between employers	0.007	0.000	0.000
within employers	0.032	0.027	0.041
Output	1.000	0.986	0.997
Worker welfare	1.000	0.996	0.992
for men	1.000	0.995	0.991
for women	1.000	0.997	0.993
Total employment	0.771	0.763	0.764
for men	0.764	0.760	0.722
for women	0.781	0.768	0.825

Notes: Table reports results from two counterfactual policy experiments. Log amenities are defined as the log amenity valuations ($\ln \bar{a} = \ln(x/w)$). Baseline results (column 0) are compared against the economy with an equal pay policy (column 1) and the economy with an equal hiring policy (column 2).

Source: Model estimates based on RAIS (2007–2014).

(c) Table 13: equal pay and equal hiring policies

6 Short summary

- The paper studies the gender pay gap using linked employer–employee data from Brazil.
- The raw gender earnings gap is 13.3 log points.
- The gender gap in employer fixed effects is 11.2 log points, equal to 84.2 percent of the overall pay gap.
- About 79.5 percent of the employer-FE gap is between employers, meaning women sort to lower-paying firms.
- The authors build an equilibrium search model with firm pay, amenities, hiring, and gender-specific firm ladders.
- Worker mobility reveals utility ranks; comparing utility ranks with pay identifies amenity valuations.
- Amenities are important for both genders, but especially important in explaining women's firm choices.
- Women have lower pay but higher amenity valuations, so the total-compensation gap is much smaller than the pay gap.
- Pay dispersion across firms mostly reflects compensating differentials rather than true utility dispersion.
- Equal pay and equal hiring policies reduce measured gender pay gaps but are not output- or welfare-improving in the estimated model.

7 Conclusion

7.1 Core conclusion

Morchio and Moser show that the gender pay gap is largely an employer-sorting phenomenon, but that sorting cannot be interpreted using pay alone. Workers value amenities, firms choose pay and amenities jointly, and men and women face different equilibrium job ladders. Once amenities are recovered from worker mobility, the gender gap in total compensation is much smaller than the pay gap.

7.2 Economic intuition

Women sort toward employers with lower pay but higher amenity valuations. This means that part of the observed pay gap is a compensating differential. However, amenities do not simply make the gender gap disappear: women still face lower total compensation, and the within-employer utility gap remains important.

Policies that force equal pay or equal hiring alter firm behavior and worker allocation, so they can close pay gaps while reducing output and welfare.

7.3 Takeaway

The paper's key lesson is that pay gaps, amenities, sorting, and equilibrium firm behavior must be studied together. A wage-only analysis overstates utility inequality across employers, but a policy analysis that ignores equilibrium adjustments overstates the benefits of equal-treatment mandates.